

Question	Answer	Marks
7(a)	• force per unit length • force per unit current • length / current perpendicular to field <i>1 mark for any two points, 2 marks for all three points.</i>	B2
7(b)(i)	$F = BQv$	B1
7(b)(ii)	arrow at Y pointing vertically upwards	B1
7(b)(iii)	upwards deflection showing circular path	B1
7(c)(i)	electric field applied vertically downwards (may be shown on a labelled diagram)	B1
	electric force on particle in opposite direction to magnetic force (may be shown on a labelled diagram)	B1
	particle undeflected when magnitudes of electric and magnetic forces are equal	B1
7(c)(ii)	$EQ = BQv$	B1
	$v = E / B$	A1

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